

BMKG STRONG MOTION ANALYZER

Laporan Analisis Strong Motion

1. RINGKASAN GUNCANGAN TERKUAT

Stasiun GKJM _ 2026-02-05 18_04_34 UTC mencatat nilai percepatan tanah maksimum (PGA) sebesar 0.633522 m/s² atau 63.352 Gal pada komponen HNN. Berdasarkan klasifikasi SIG-BMKG yang digunakan oleh BSMA, hasil tersebut termasuk Skala II (DIRASAKAN) dengan rentang intensitas III-V.

Network	IA
Station	GKJM
Location	00
Sampling Rate	100.000 Hz
Start Time	2026-02-05T18:04:34.820000Z
End Time	2026-02-05T18:12:34.810000Z
Channels	HNE, HNN, HNZ

SKALA II SIG-BMKG | DIRASAKAN

Warna: Hijau | Rentang intensitas: III-V

Dampak: Dirasakan oleh orang banyak tetapi tidak menimbulkan kerusakan. Benda-benda ringan yang digantung bergoyang dan jendela kaca bergetar.

INSTRUMENTAL INTENSITY (MMI): VI | PERCEIVED SHAKING: STRONG

Potensi Kerusakan: Light | Peak Acc Baseline: 12 %g | Peak Vel Baseline: 9.6 cm/s

Nilai Rekaman Terukur: PGA = 6.460 %g | PGV = 5.243 cm/s (Skala USGS ShakeMap / Worden et al., 2011)

2. PARAMETER STRONG MOTION

Channel	PGA (Gal)	PGV (cm/s)	PGD (cm)	Arias (m/s)	D5-95 (s)	SIG
HNE	47.400	3.4188	0.5096	0.020685	11.32	II (III-V)
HNN	63.352	5.2428	0.5622	0.032571	8.27	II (III-V)
HNZ	23.526	1.7336	0.1792	0.008285	27.49	II (III-V)

Parameter komponen terkuat:

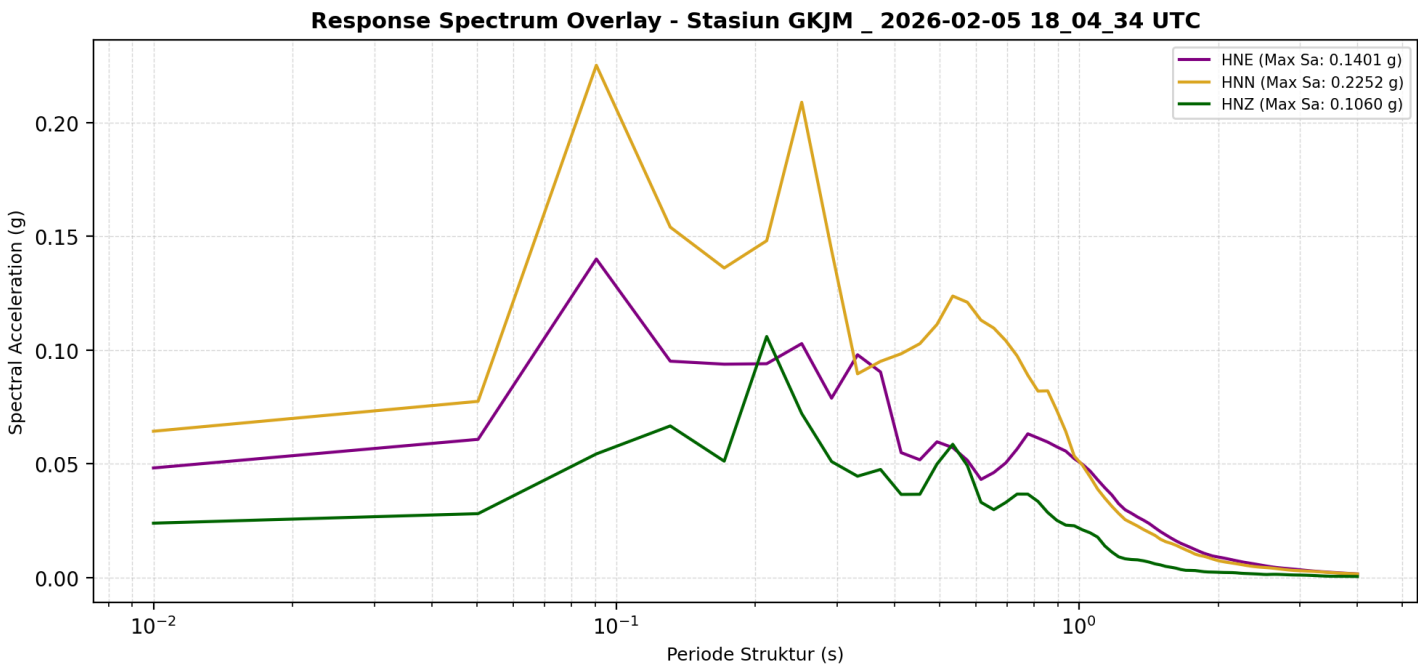
PGA = 0.633522 m/s² (63.352 Gal); PGV = 5.2428 cm/s; PGD = 0.5622 cm; Arias Intensity = 0.032571 m/s; D5-95 = 8.27 s.

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3. RESPONSE SPECTRUM

Kurva response spectrum menunjukkan percepatan spektral sebagai fungsi periode struktur berdasarkan hasil pemrosesan response spectrum pada pipeline BSMA.

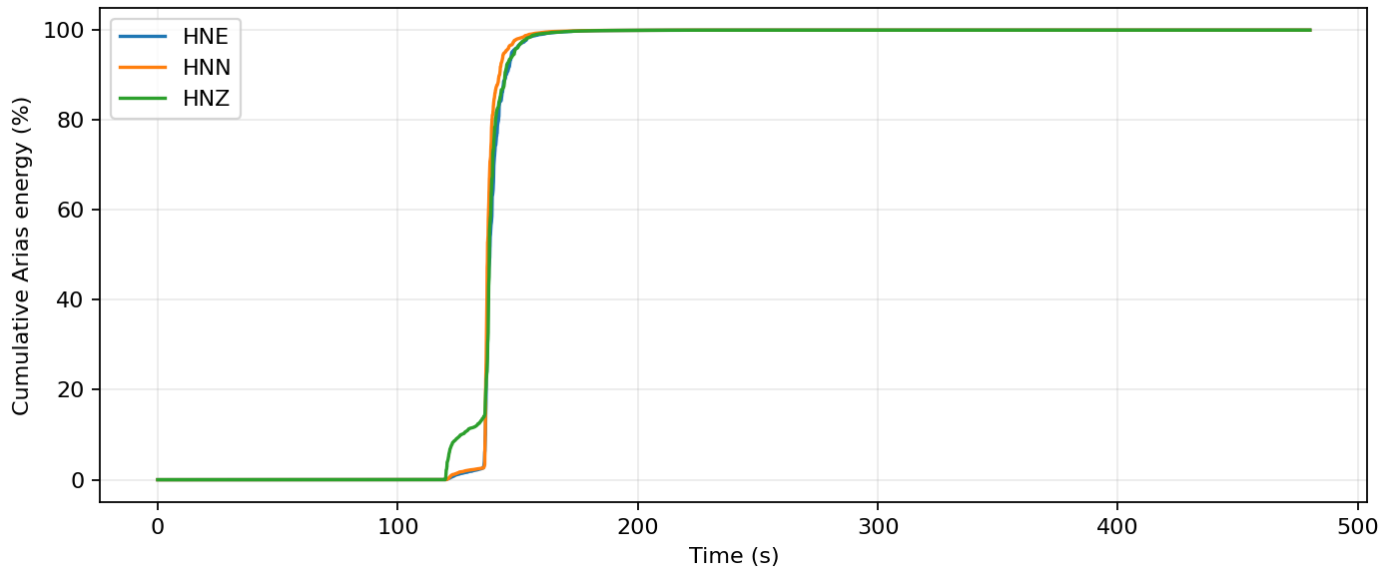


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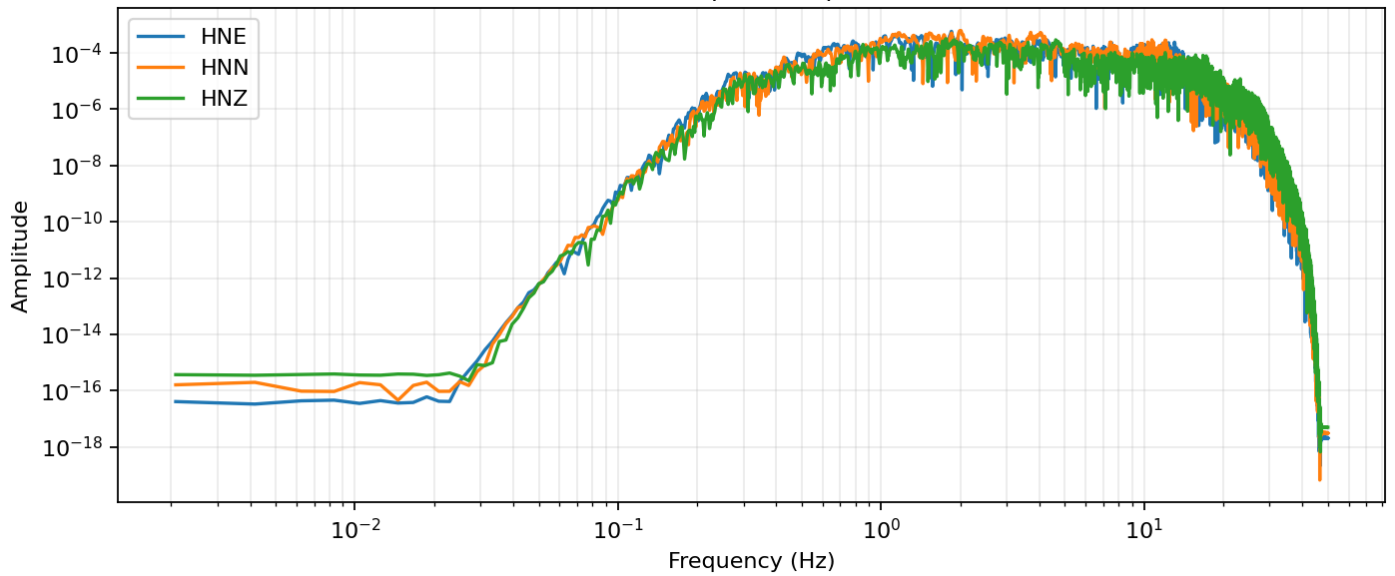
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4. HUSID PLOT DAN FOURIER AMPLITUDE SPECTRUM

Husid energy curve



Fourier amplitude spectrum



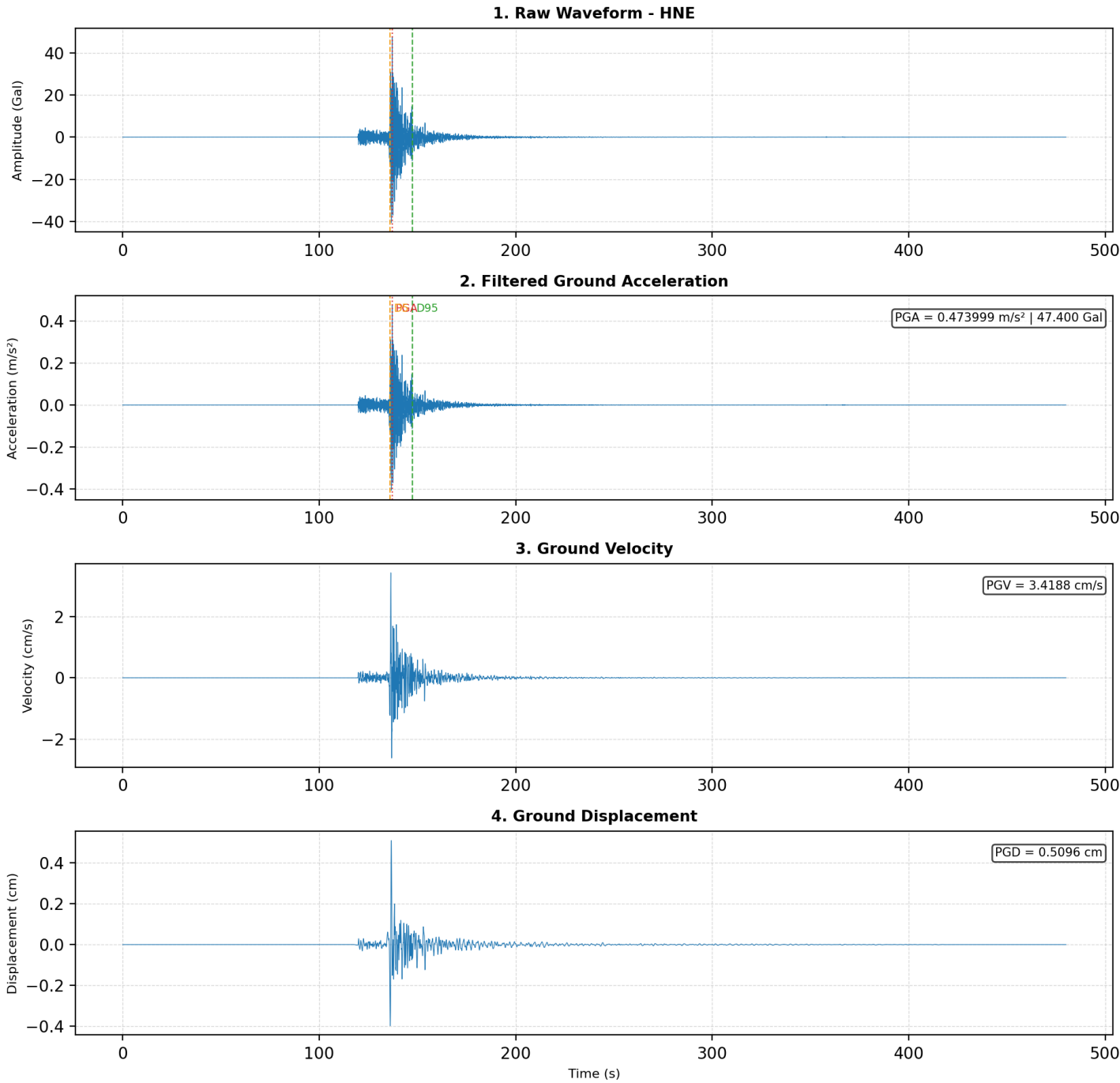
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5. WAVEFORM ANALYSIS - HNE

PGA: 47.400 Gal | PGV: 3.4188 cm/s | PGD: 0.5096 cm | D5-95: 11.32 s

Analisis Kinematik Ground Motion
Stasiun: GKJM _ 2026-02-05 18_04_34 UTC | Channel: HNE



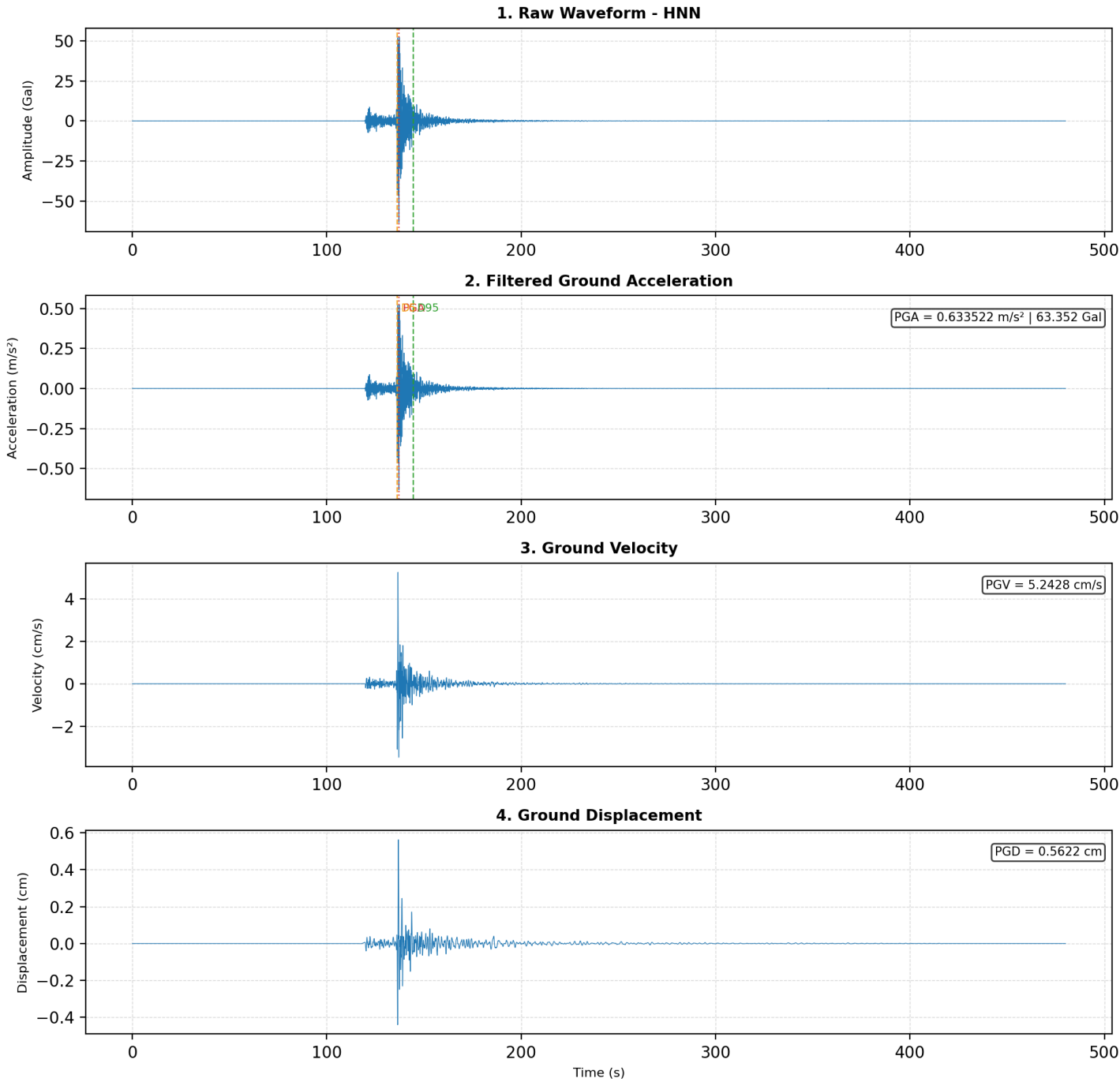
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5. WAVEFORM ANALYSIS - HNN

PGA: 63.352 Gal | PGV: 5.2428 cm/s | PGD: 0.5622 cm | D5-95: 8.27 s

Analisis Kinematik Ground Motion
Stasiun: GKJM _ 2026-02-05 18_04_34 UTC | Channel: HNN



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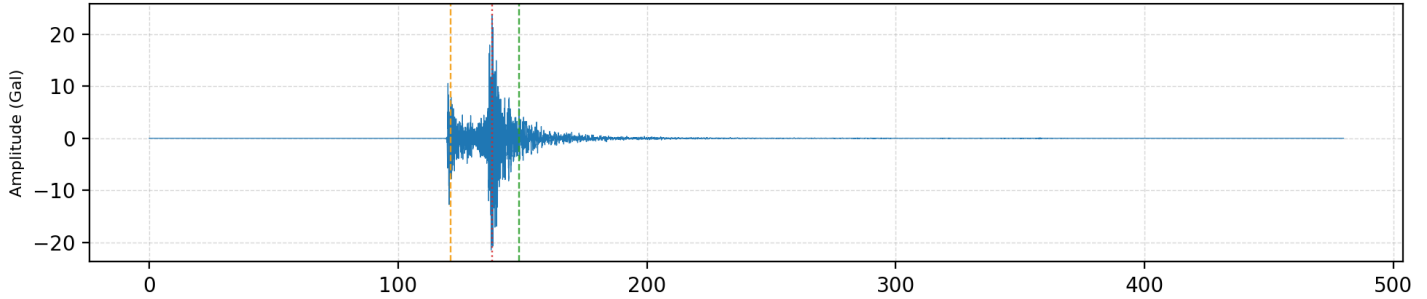
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5. WAVEFORM ANALYSIS - HNZ

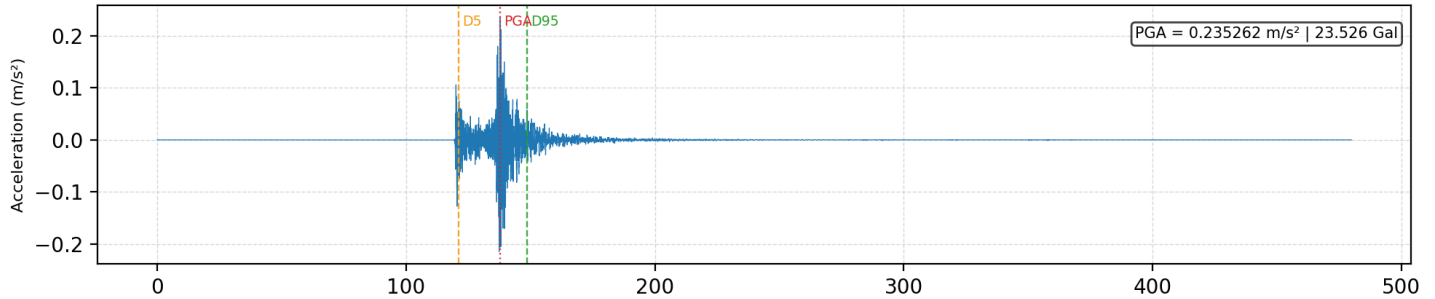
PGA: 23.526 Gal | PGV: 1.7336 cm/s | PGD: 0.1792 cm | D5-95: 27.49 s

Analisis Kinematik Ground Motion
Stasiun: GKJM _ 2026-02-05 18_04_34 UTC | Channel: HNZ

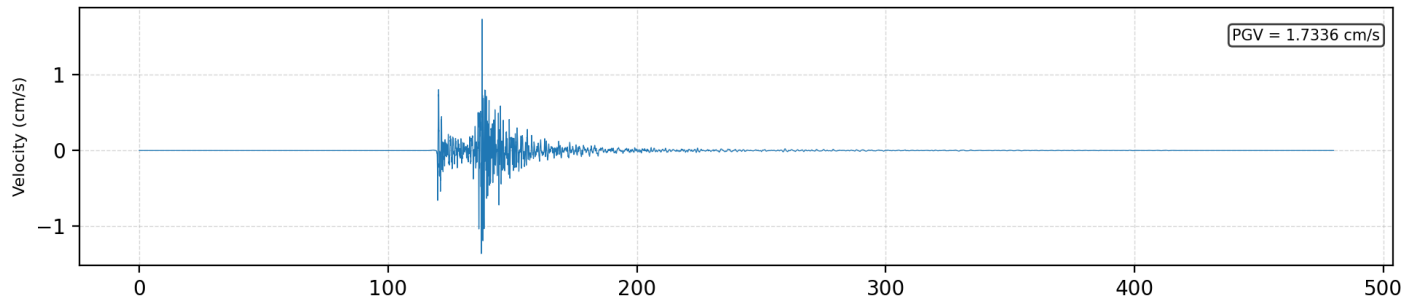
1. Raw Waveform - HNZ



2. Filtered Ground Acceleration



3. Ground Velocity



4. Ground Displacement

